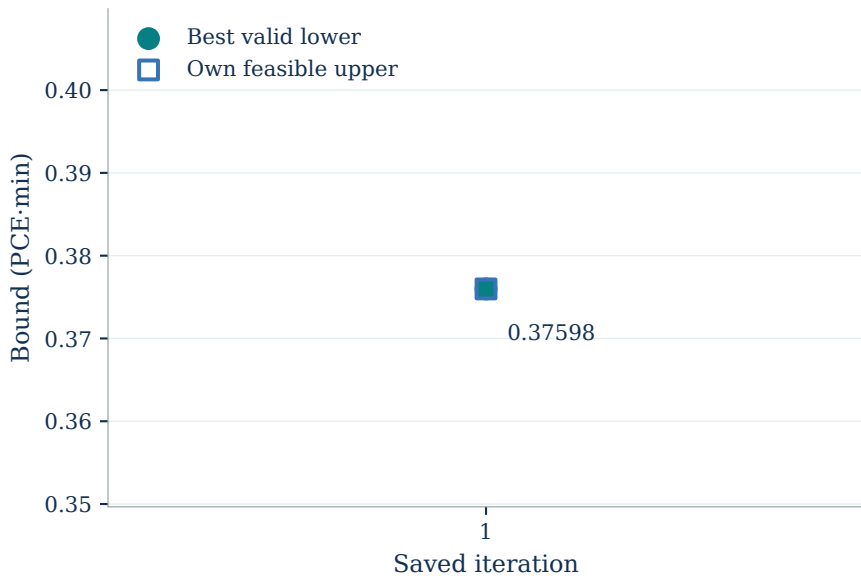


Lagrangian bounds and certified gap

a Separate lower and upper bounds



b Same-instance certificate

